

Technology Stack

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	Streamlit, HTML, CSS	Streamlit was chosen to rapidly build an interactive web interface for entering problem descriptions and viewing emotion-aware results. Custom HTML and CSS enhance the UI and improve user experience.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, PyTorch	Python and PyTorch run the BiLSTM and BERT emotion classification models and orchestrate calls to the Gemini API for generating personalized guidance.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV Files (interactions.csv)	CSV files were used for lightweight local logging of student interactions, predicted emotions, and responses, without requiring a dedicated database server.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Local Deployment using Streamlit	The application is currently deployed locally for development and demonstration purposes. The architecture can be migrated to cloud platforms in future versions.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git, GitHub	Git and GitHub were used for source code management, version tracking, collaboration, and project backup.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Gemini AI, Hugging Face Transformers, PyTorch, NumPy	Gemini AI generates personalized tips and encouragement, Hugging Face Transformers and PyTorch power the BiLSTM/BERT emotion classifiers, and NumPy supports numerical data processing.